

9. Find the period of the function $f(x) = |\sin x|$. Draw its graph. Indicate which type of waveform is this.

10. Show that the function $f(x) = x$, $0 < x < 2$ gives a saw-toothed wave form.

11. Find whether the following function is even or odd :

(i) $f(x) = 2$, $0 < x < 3$
 $= -2$, $-3 < x < 0$.

(ii) $f(x) = \cos x$, $0 < x < \pi$
 $= 0$, $\pi < x < 2\pi$.

(iii) $f(x) = x(10 - x)$, $0 < x < 10$.

✓ 12. Show that every function is a sum of a even function and an odd function. [W.B.U.T. 2005, 2013]

[Hint : Express $f(x) = \frac{1}{2}\{f(x) + f(-x)\} + \frac{1}{2}\{f(x) - f(-x)\}$
 $= \phi(x) + \psi(x)$.]

13. Show that for $-\pi < x < \pi$,

$$x \cos x = -\frac{1}{2} \sin x + 2 \left(\frac{2}{1.3} \sin 2x - \frac{3}{2.4} \sin 3x + \frac{4}{3.5} \sin 4x - \dots \right)$$

14. Expand the Fourier series for $f(x) = x$ over $-\pi \leq x \leq \pi$.

Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \dots = \frac{\pi}{4}$. [W.B.U.T. 2012]

✓ 15. Find the Fourier series corresponding to the function

$$f(x) = \pi - x, \quad 0 < x < \pi$$

$$= x - \pi, \quad -\pi < x < 0.$$

What is the value of the series at $x = 0$ and at $x = \pi$.

✓ 16. Obtain the Fourier series of the function

$$f(x) = 0, \quad -\pi < x \leq 0$$

$$= \pi, \quad 0 < x \leq \pi.$$

Find whether the series is convergent.

✓ 17. Find the Fourier series of the function

$$f(x) = 0, \quad -2 < x < 0$$

$$= 1, \quad 0 < x < 2 \text{ and } f(x+4) = f(x).$$

Find $f(0)$, $f(-2)$ and $f(2)$ so that the Fourier series converges to $f(x)$ for $-2 \leq x \leq 2$.

18. Find the Fourier expansion of x^2 on $[-\pi, \pi]$.

Hence find the value of

(i) $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

~~(ii) $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$~~

[W.B.U.T. 2005, 2012]

(iii) $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

19. Prove that $x + x^2 = \frac{\pi^2}{3} - 4 \left\{ \frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \dots \right\}$

$$+ 2 \left\{ \frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \dots \right\}$$

Hence find the value of $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

20. Find the Fourier series for the function defined by

$$f(x) = 1, \quad 0 < x < \pi$$

$$= 0, \quad x = 0$$

$$= -1, \quad -\pi < x < 0.$$

Tell whether the series is an expansion of $f(x)$.

Hence deduce the value of the series, $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

21. Find the Fourier series of the function

$$f(x) = -k, \quad -\pi < x < 0$$

$$= k, \quad 0 < x < \pi \quad \text{and} \quad f(x + 2\pi) = f(x).$$

22. Find the Fourier series of the function

$$f(x) = 0, \quad -2 < x < -1$$

$$= k, \quad -1 < x < 1$$

$$= 0, \quad 1 < x < 2.$$

Draw the graph of the extended periodic function $f(x)$ of period 4. Mention the Waveform given.

23. A sinusoidal voltage $E \sin \omega t$ is passed through a half-wave rectifier which clips the negative portion of the wave. Find the Fourier series of the resulting periodic function.

$$f(t) = 0, \quad -\frac{T}{2} < t < 0$$

$$= E \sin \omega t, \quad 0 < t < \frac{T}{2}, \quad \text{where } T = \frac{2\pi}{\omega}.$$

24. Find the waveform given by the function $f(x) = x + \pi$, $-\pi < x < \pi$ and $f(x + 2\pi) = f(x)$. Obtain its Fourier series.

25. Find the Fourier expansion for $f(x)$,

✓ if $f(x) = -\pi, \quad -\pi < x < 0$

$$= x, \quad 0 < x < \pi.$$

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}.$

[Hint : at $x = 0$ the Four. series $= \frac{1}{2} \left\{ \lim_{t \rightarrow 0^-} f(t) + \lim_{t \rightarrow 0^+} f(t) \right\} = -\frac{\pi}{2}$]

26. Expand $f(x) = e^{-x}$ as a Fourier series in the interval $(-4, 4)$.

✓ 27. Find the Fourier series for the function

$$f(x) = 1 + \frac{2x}{\pi}, \quad -\pi \leq x \leq 0$$

$$= 1 - \frac{2x}{\pi}, \quad 0 \leq x \leq \pi.$$

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}.$

28. Find the Fourier series corresponding to the function

$$f(x) = -x, \quad -\pi < x < 0$$

$$= 0, \quad 0 < x < \pi.$$

29. Find the Fourier series of the function $|\sin x|$, $-\pi < x \leq \pi$. Find whether this Fourier series is a Fourier expansion of $|\sin x|$.

30. Let $f(x) = x$, $0 \leq x \leq \frac{\pi}{2}$

$$= \pi - x, \quad \frac{\pi}{2} \leq x \leq \pi$$

$$= -f(-x), \quad -\pi \leq x \leq 0.$$

Show that $f(x)$ satisfies Dirichlet's condition on $[-\pi, \pi]$. Obtain the Fourier series for f in $[-\pi, \pi]$.

31. Find the Fourier series of the function

$$f(x) = -\frac{\pi}{4}, \quad -\pi \leq x \leq 0$$

$$= \frac{\pi}{4}, \quad 0 < x \leq \pi.$$

32. Find the Fourier series of the function

$$f(x) = 0, \quad -\pi < x < 0$$

$$= \frac{1}{2}, \quad x = 0$$

$$= 1, \quad 0 < x < \pi.$$

Does it converge at $x = 0$. Find the value at this point.

33. Find the Fourier series corresponding to the function

$$f(x) = x, \quad -\pi \leq x \leq 0$$

$$= 2x, \quad 0 \leq x \leq \pi.$$

Does the series equals $f(x)$ everywhere ?

34. Draw the graph of the following periodic function of period
- 2π
- . Identity the waveform given by it. Expand the function into its Fourier expansion :

$$f(x) = 0, \quad -\pi \leq x \leq 0$$

$$= \sin x, \quad 0 < x \leq \pi.$$

Hence show that $\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots \infty = \frac{1}{4}(\pi - 2)$.

35. Prove that
- $f(x) = \frac{\pi^2}{16} + \sum_{n=1}^{\infty} \frac{1}{4n^2} (\cos n\pi - 1) \cos nx - \sum_{n=1}^{\infty} \frac{\pi}{4n} \cos n\pi \sin nx$
-
- when

$$f(x) = 0, \quad -\pi < x \leq 0$$

$$= \frac{\pi x}{4}, \quad 0 < x < \pi.$$

Hence deduce that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots = \frac{\pi^2}{8}$.

36. Find the Fourier expansion of the following function
- f
- of period
- 2π
- defined as

$$f(x) = 0, \quad -\pi \leq x \leq 0$$

$$= x, \quad 0 < x \leq \pi.$$

Find the sum of the series at $x = -5\pi$. Hence deduce that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}.$$

37. Show that on $-\pi \leq x \leq \pi$,

$$\frac{1}{2}\pi - x \sim \frac{1}{2}\pi - 2\left(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \dots\right).$$

38. Draw the graph of the function $f(x) = 0, -\pi < x < 0$
 $= x^2, 0 < x < \pi$.

If $f(2\pi + x) = f(x)$, obtain Fourier series of $f(x)$.

39. Find the Fourier series of the function

$$f(x) = x^2, \quad 0 \leq x \leq \pi$$

$$= -x^2, \quad -\pi \leq x \leq 0.$$

40. Find a Fourier expansion for $f(t) = 1 - t^2, -1 \leq t \leq 1$.

41. Expand the function $\sin ax$ in Fourier series in $(-\pi, \pi)$.

✓ 42. Find the Fourier series of $|x|, -\pi < x < \pi$.

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots = \frac{\pi^2}{8}$. [W.B.U.T. 2014]

43. Expand $f(x) = |\cos x|$ as a Fourier series in $(-\pi, \pi)$. Does it converge to $f(x)$ for all values of x . Give reasons.

✓ 44. If $f(x) = x + x^2, -\pi < x < \pi$, be represented in Fourier series as $\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$ then find a_0 . [W.B.U.T. 2004, 2012]

✓ 45. A function $f(x) = x^2, -\pi < x \leq \pi$ is represented by a Fourier Series as $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$. Find a_0 and show that $b_n = 0$ for all n . [W.B.U.T. 2005]

46. Expand $x \sin x, 0 < x < 2\pi$ in a Fourier series.

47. Expand $f(x) = x^2, 0 < x < 2\pi$ in a Fourier series. Hence find the value of $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$.

48. $f(x)$ with period 2π is defined by the relation

$$f(x) = \frac{\pi - x}{2}, \quad 0 < x < 2\pi$$

$$= 0, \quad x = 0.$$

Find its Fourier series.

49. Express $f(x) = \pi - x$ in a cosine series in $0 < x < \pi$. Prove that the sum of the series for $x = \pm\pi$ is 0. Hence deduce the sum $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$ [W.B.U.T. 2013]

50. Given $f(x) = \frac{1}{4}\pi x$, $0 \leq x \leq \frac{\pi}{2}$
 $= \frac{1}{4}\pi(\pi - x)$, $\frac{\pi}{2} \leq x \leq \pi$.

Expand $f(x)$ in an infinite series of

(i) Cosines of multiples of x in $[0, \pi]$.

(ii) Sines of multiples of x .

51. Expand $f(x) = x$, $0 < x < 2$ in a half range cosine series.

52. Find the half Range sine expansion of the function

$$f(x) = \frac{2k}{l}x, \quad 0 < x < \frac{l}{2}$$

$$= \frac{2k}{l}(l - x), \quad \frac{l}{2} < x < l.$$

Mention its waveform also.

53. Find the Half Range Cosine series of the function.

$$f(x) = \frac{\pi}{3}, \quad 0 \leq x < \frac{\pi}{3}$$

$$= 0, \quad \frac{\pi}{3} < x < \frac{2\pi}{3}$$

$$= -\frac{\pi}{3}, \quad \frac{2\pi}{3} < x \leq \pi$$

$$= \frac{\pi}{6}, \quad x = \frac{\pi}{3}$$

$$= -\frac{\pi}{6}, \quad x = \frac{2\pi}{3}$$

Does the series equal $f(x)$ everywhere?

54. Prove that if c be a constant then

$$c = \frac{4}{\pi}c \left(\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \dots \right), \quad 0 < x < \pi.$$

55. Show that on $0 < x < \pi$,

$$x^2 = \frac{2}{\pi} \left\{ \left(\frac{\pi^2}{1} - \frac{4}{1^3} \right) \sin x - \frac{\pi^2}{2} \sin 2x + \left(\frac{\pi^2}{3} - \frac{4}{3^3} \right) \sin 3x - \dots \right\}.$$

56. Expand $f(x) = \cos x$, $0 \leq x < \pi$ in Fourier sine series in $0 \leq x \leq \pi$.

57. Find the Half Range sine series of the function

$$f(x) = 0, \quad 0 \leq x < \frac{\pi}{2}$$

$$= \frac{\pi}{2}, \quad \frac{\pi}{2} < x < \pi.$$

Define $f\left(\frac{\pi}{2}\right)$ and $f(\pi)$ so that the obtained series equals $f(x)$ everywhere.

58. Find the Half Range sine series of the function defined by

$$f(x) = \frac{1}{3}\pi, \quad 0 < x < \frac{\pi}{3}$$

$$= 0, \quad \frac{\pi}{3} < x < \frac{2\pi}{3}$$

$$= -\frac{\pi}{3}, \quad \frac{2\pi}{3} < x < \pi.$$

Find $f(0)$, $f(\pi)$, $f\left(\frac{\pi}{3}\right)$, $f\left(\frac{2\pi}{3}\right)$ so that the series converges to $f(x)$ for all values of x .

59. Prove that $\frac{\pi}{2} - x = \frac{4}{\pi} \left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right)$, $0 \leq x \leq \pi$.

60. Prove that the function $f(x) = \frac{1}{2}k - x$, $0 < x < k$ is represented by the series $\frac{k}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{2n\pi x}{k}$.

61. If $f(x) = \frac{(\pi - x)^2}{4}$ in $0 \leq x \leq 2\pi$, show that $f(x) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$.

62. Find the Fourier series to represent the function $f(x)$ given by

$$f(x) = x, \quad 0 \leq x \leq \pi$$

$$= 2\pi - x, \quad \pi \leq x \leq 2\pi.$$

Hence find the value of $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

63. Obtain the Fourier series for $f(x) = \pi x$ in $[0, 2]$.

64. Find the Half-Range cosine series for the function $f(x) = (x-1)^2$ in the interval $0 < x < 1$.

Hence find the value of $8\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right)$.

65. Find the half-range sine series for the function $f(t) = t - t^2$, $0 < t < 1$.

66. Find the half range sine series of e^x in $0 < x < 1$.

✓ 67. Find a half-range cosine series for

$$f(x) = ax, \quad 0 \leq x \leq l/2$$

$$= a(l-x), \quad \frac{l}{2} \leq x \leq l.$$

Hence find the value of $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$.

✓ 74. Prove that for $0 \leq x \leq \pi$

$$x(\pi - x) = \frac{\pi^2}{6} - \left(\frac{\cos 2x}{1^2} + \frac{\cos 4x}{2^2} + \frac{\cos 6x}{3^2} + \dots \right).$$

Using Parseval's Identity show that $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$.

✓ 75. Prove that for $0 \leq x \leq \pi$

$$x(\pi - x) = \frac{8}{\pi} \left(\frac{\sin x}{1^3} + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \dots \right).$$

Using Parseval's identity show that $\sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}$.